

Review article

Association between benzodiazepines and related drug use and venous thromboembolism risk: A systematic review and meta-analysis

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ABSTRACT

Benzodiazepines and related drugs (BZDRs) are commonly prescribed for insomnia, anxiety, and other psychiatric disorders. However, the association between BZDR use and venous thromboembolism (VTE) remains controversial. This meta-analysis evaluated this association systematically. We searched PubMed, Embase, and the Cochrane Library from inception to February 2025 for relevant case-control and cohort studies. Random- or fixed-effect models were used to estimate the overall relative risk. Eight studies met the eligibility criteria and were included in the analysis. BZDR exposure was associated with a significantly increased risk of VTE (OR 1.47, 95 % CI 1.19–1.7; $p < 0.001$; $I^2 = 88\%$; 8 studies with 10 estimates). Subgroup and sensitivity analyses revealed positive associations. Significant statistical and clinical heterogeneity was observed in the main and most subgroup analyses. These findings suggest an increased risk of VTE in BZDR users, emphasizing the need for careful consideration when prescribing BZDRs to patients at risk of VTE. Given the few studies included, well-designed prospective studies controlling for important confounders are needed to verify our findings.

1. Introduction

Venous thromboembolism (VTE) is characterized by the formation of blood clots in the venous system, most commonly presenting as deep venous thrombosis (DVT) or pulmonary embolism (PE) [1]. VTE is associated with significant morbidity, frequent recurrence, and substantial healthcare costs [2]. In the United States, VTE is responsible for approximately 6000–10,000 deaths annually [3]. However, most of these complications and deaths are preventable with appropriate interventions. Despite its relatively low prevalence, VTE affects more than one million people in Europe each year, and nearly 10 million people worldwide [4]. In addition to advancing age, major risk factors for thrombosis include physical inactivity, obesity, trauma, surgery, malignancy, pregnancy, and oestrogen-related medications [5].

Benzodiazepine and related drugs (BZDRs) are the most frequently prescribed medications for insomnia and anxiety worldwide [6]. The overall annual consumption of BZDRs has been increasing in many

countries [7], especially in countries with a burgeoning gross domestic product [8–10]; however, a recent study showed a 1.88 % annual decreased consumption of BZDs but a 3.28 % increased consumption of Z-drugs from 2008 to 2018 [11]. Although BZDRs are generally effective and well-tolerated, recent studies suggest potential adverse effects, including an increased risk of falls [12], bone fractures [13], infections [14], and cognitive decline [15]. Recently, the association between BZDR use and VTE has garnered increasing attention; however, relevant studies have yielded inconsistent results. For example, Alenka et al. [16] reported no increased VTE risk associated with BZDR use in postsurgical patients in a case-control study with a total case of 171. Conversely, another study including 337 cases found an association between benzodiazepine use and increased VTE risk in pregnant women [17]. Furthermore, two studies [18,19] evaluating VTE risk in psychiatric unit patients yielded conflicting findings; however, these two studies were limited by the sample size. Since then, several population-based studies explored this relationship were published. In a case-control study, Chen

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et al. [20] reported that BZD use was associated with an increased risk of VTE and their findings were confirmed by two other studies [21,22]; however, in another study [23], the use of Z-drugs was found not to be associated with an increased risk of VTE. Thus, the relationship between BZDRs exposure and the risk of VTE is still unclear.

Given the high mortality of VTE cases and wide use of BZDR medication, we conducted a systematic literature review and meta-analysis to evaluate the relationship between BZDR use and VTE risk with the dual aim of facilitating the development of evidence-based practice guidelines and informing future research.

2. Methods

2.1. Literature search

Our systematic review and meta-analysis were conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Table S1). This meta-analysis was not pre-registered or pre-protocolled. A systematic literature search was performed in PubMed, Embase, and the Cochrane Library from inception to 1 January 2025, using predefined keywords. We used the keywords [hypnotic OR benzodiazepine OR benzodiazepines OR chlordiazepoxide OR clonazepam OR alprazolam OR diazepam OR bromazepam OR nitrazepam OR estazolam OR lorazepam OR flurazepam OR flunitrazepam OR midazolam OR prazepam OR triazolam OR temazepam OR oxazepam OR medazepam OR phenazepam OR fenazepam OR cloxazolam OR oxazolam OR quazepam OR clobazam OR nordazepam OR z-drug OR zolpidem OR zopiclone OR zaleplon OR eszopiclone] AND [venous thrombosis OR venous thromboembolism OR deep vein thrombosis OR pulmonary embolism] for our search. In addition, we conducted a manual search of the reference lists of retrieved articles, reviews, and meta-analyses to identify additional relevant studies.

2.2. Inclusion criteria

Two reviewers (JKZ and WY) independently screened the titles and abstracts of all identified articles to determine their eligibility. Discrepancies were resolved through discussion with a third reviewer (DH or XFX). Studies were included if they met all of the following criteria: randomised controlled trials (RCTs), cohort, nested case-control, and case-control studies; exposure, patients exposed to BZDRs; comparison, a reference group of patients not exposed to BZDRs; outcome, reported odds ratios (ORs), relative risks, or hazard ratios for VTE; and sufficient data to calculate risk estimates. We excluded case reports, case series, animal studies, editorials, conference abstracts, and review articles.

2.3. Data extraction and quality assessment

Data were extracted independently by two reviewers in a double-blinded manner, with any discrepancies resolved by a third reviewer. Extracted data included the first author's name, publication year, study design, study setting, study location, BZDR medication assessment method, drug indication, outcome measures, number of cases and controls, follow-up duration, and study quality. When multiple effect size estimates were reported, the most adjusted estimate was used. Methodological quality was assessed using the Newcastle–Ottawa Scale for observational studies [24], as recommended by the Cochrane Collaboration. The risks of selection, comparability, and exposure/outcome assessment bias were evaluated by awarding stars to each of nine items. The scale features eight criteria and yields scores ranging from 0 (high risk of bias) to 9 (low risk of bias). Studies with scores ≥ 7 were considered to be of high quality.

2.4. Outcomes assessment

The association between BZDR use and VTE risk was evaluated using adjusted analyses. Sensitivity analysis was performed by sequentially omitting individual studies to assess their influence on the overall risk estimate. In addition, subgroup analyses were conducted based on study quality (high vs. low), the number of adjusted variables (≥ 5 vs. < 5), type of agent (BDZs vs. Z-drugs), study design (case–control vs. cohort), drug indication, or patient characteristics.

2.5. Data analysis

Statistical analyses were performed using Stata SE software (version 10.0; Stata Corp, College Station, TX, USA). Heterogeneity across studies was assessed using the I^2 statistic, with values of 0–25 %, 25–50 %, 50–75 %, and > 75 % indicating very low, low, moderate, and high heterogeneity, respectively [25]. Given the clinical heterogeneity observed among the included studies, we employed a random-effects model for all data pooling [26]. Publication bias was assessed using Egger's test [26]. P -values < 0.05 were considered statistically significant.

3. Results

3.1. Search results

Our literature search initially identified 1023 citations, reduced to 851 after removing duplicates. Following title and abstract screening, 829 articles were excluded, leaving 22 studies for full-text assessment. Two studies [20,21] used the Longitudinal Health Insurance Database in Taiwan; however, both were included in the meta-analysis as these analyses will use a random sample and the degree of overlap is likely even less than might be assumed based overlap years as these may be separately derived samples. However, two studies conducted by Richardson et al. [23,27] used Clinical Practice Research Datalink in England. Because of overlapping study periods, only one study [23] was included in our meta-analysis. Ultimately, eight studies were included in the meta-analysis, comprising approximately 18,997 VTE cases from five case–control studies and three cohort studies. The number of articles excluded at each stage of the eligibility assessment is detailed in Fig. 1.

3.2. Characteristics of included studies

Table 1 presents the baseline characteristics of the included studies. Publication dates ranged from 2012 to 2024. Five studies used population-based databases, including two from the same nationwide claims database in Taiwan. The study populations were predominantly Western ($n = 5$), with three studies conducted in Asian populations. Considerable heterogeneity existed in the studied populations, encompassing general populations, patients with dementia, trauma patients, and individuals admitted to psychiatric units. Regarding BZDR medications, five studies focused exclusively on benzodiazepines, two examined benzodiazepines and Z-drugs, and one investigated Z-drugs. Among the included studies, three studies did not adjust for any potential confounding and two adjusted for several important potential confounding not including indications. However, the remaining studies addressed the indications by statistical adjustment or disease risk score matched methods. The extent of adjustment for potential clinical risk factors varied across studies.

The quality scores of the included studies ranged from 5 to 9, with five studies classified as high quality and three as low quality. The mean quality score was 7.25, with detailed scoring information provided in Tables S2 and S3.

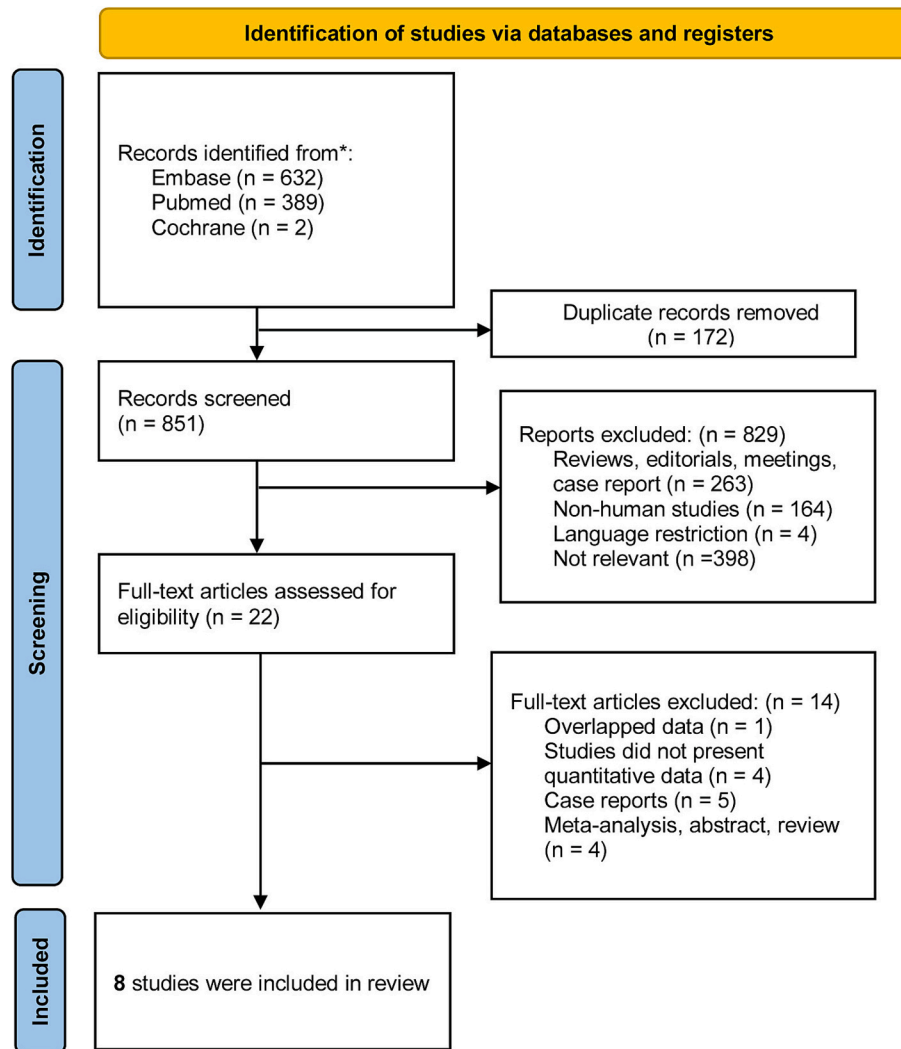


Fig. 1. Flow diagram of study selection process.

3.3. Meta-analysis

Table 2 presents the results of all analyses. Our meta-analysis of eight studies (10 estimates) demonstrated a statistically significant association between BZDR use and increased VTE risk (OR 1.42, 95 % confidence interval [CI] 1.19–1.7; $p < 0.001$) (Fig. 2). High heterogeneity was observed across the studies ($I^2 = 88\%$). As shown in Fig. S1, we did not observe any evidence of publication bias (Egger test, $P = 0.92$). Sensitivity analysis demonstrated no significant change in the pooled risk estimates upon the exclusion of any single study, with pooled ORs for VTE ranging from 1.38 to 1.52.

Subgroup analysis based on study quality revealed a significant association between BZDR use and VTE in both high-quality studies (OR 1.47, 95 % CI 1.14–1.9; $p = 0.003$; $I^2 = 91\%$; 4 studies with 5 estimates) and low-quality studies (OR 1.27, 95 % CI 1.1–1.45; $p = 0.001$; $I^2 = 32.7\%$; 4 studies with 5 estimates).

Subgroup analysis by study setting showed a significant association between BZDR use and VTE in population-based studies (OR 1.38, 95 % CI 1.14–1.68; $p = 0.001$; $I^2 = 91.7\%$; 4 studies with 5 estimates). In contrast, the pooled OR of hospital-based studies was high but fell just short of statistical significance (OR 1.72, 95 % CI 0.99–3.01; $p = 0.051$; $I^2 = 32.2\%$; 3 studies with 3 estimates).

Subgroup analysis by study design demonstrated a significant association between BZDR use and VTE risk in the cohort (OR: 1.23, 95 % CI: 1.11–1.36; $p < 0.001$; $I^2 = 13.2\%$; 4 studies with 5 estimate) and

case-control (OR: 1.53, 95 % CI: 1.19–1.97; $P = 0.001$; $I^2 = 90.8\%$; 4 studies with 5 estimates) studies. Regarding the number of adjustment variables, studies that adjusted for at least five variables demonstrated a significantly increased VTE risk (OR: 1.47, 95 % CI: 1.14–1.90; $P = 0.003$; $I^2 = 91\%$; 3 studies with 3 estimates). In contrast, the pooled OR of studies adjusting for fewer than five variables was high but fell just short of statistical significance (OR: 1.72, 95 % CI: 0.99–3.01; $P = 0.051$; $I^2 = 32.2\%$; 5 studies with 7 estimates). Further analysis limited to three studies adjusting for indication demonstrated an increased risk of VTE in patients exposed to BZDR (OR: 1.42, 95 % CI: 1.11–1.84; $P = 0.008$; $I^2 = 92.9\%$; 3 studies with 4 estimates).

When the studies were grouped based on BZDR type, a statistically significant positive association with VTE risk was observed for BZD (OR 1.56, 95 % CI 1.26–1.92; $p < 0.001$; $I^2 = 85.41\%$; 6 studies with 8 estimates) and Z-drug (OR 1.33, 95 % CI 1.05–1.68; $p = 0.017$; $I^2 = 0\%$; 2 studies with 3 estimates).

Two studies evaluated a potential dose–response relationship. Shih et al. divided patients with BZDR prescription into tertiles (low, medium, high) by the cumulative dose (DDD), identifying a dose-dependent relationship between BZDR use and DVT risk. Meanwhile, Chen et al. demonstrated that the odds of VTE occurrence increased with higher DDD and number of BZDR prescriptions.

Table 1
Characteristics of the included studies.

Study, year	Location, setting	Study design	Study period	Population	Exposure measurement	Type of BZRD	Definition of DVT	Total population	Adjustment	Quality
Delluc et al. [18],	France, hospital-based	Cohort	2006–2010	Patient admitted psychiatric units	Medical record review	Benzodiazepine	ICD-10	Exposure 11/285 No-exposure 5/173	NA	5
Jensen et al. [17],	Denmark, hospital-based	Case-control	2003–2010	Pregnant women	Danish pharmacies	Benzodiazepine	ICD-10	Case 337 Control 299,473	Age, BMI, smoking, previous VTE, previous stroke, vascular disease, thrombophilia, gestational hypertension, hyperemesis, gestational diabetes, pre-eclampsia, and calendar time period	9
Premuš Marušić et al. [16]	Slovenia, hospital-based	Case-control	2007–2011	Surgical patients	Medical record review	Benzodiazepine and Z-drugs	Medical record review	Case 144 Control 142	NA	5
Chen et al. [20]	China, population-based	nested case-control	2002–2012	General population	ATC classification system	Benzodiazepine and Z-drugs	ICD-9-CM	Case 2800 Control 2800	Disease risk score, Charlson Comorbidity Index score (0, 1–2, ≥3)	8
Richardson et al. [23],	England, population-based	Cohort	2000–2016	General population with dementia	Clinical Practice Research Datalink	Z-drugs	ICD-10	Exposure 22/3532 No-exposure 43/10,214	Gender, age, ethnicity, current smoker, alcohol drinker, BMI, systolic blood pressure, dementia subtype and proxies for dementia severity, osteoporosis, other musculoskeletal conditions, depression, depression symptoms, anxiety, anxiety symptoms, Parkinson's disease, urinary incontinence, age related macular degeneration, glaucoma, cataract, other visual impairment, diabetes, hyperlipidaemia, hypertension, heart attack, heart failure, atrial fibrillation, ischaemic stroke/TIA, angina, venous thromboembolism, osteoarthritis, rheumatoid arthritis, migraine/headache, back/neck pain, cancer, and Chronic Obstructive Pulmonary Disease, concurrent medication use	9
Skicki et al. [22],	USA, population-based	Cohort	2013–2018	Surgical patients	Urine drug screen	Benzodiazepine	Pennsylvania Trauma Outcome Study database	Exposure 946/ 33,260 No-exposure 2102/ 130,013	Age, sex, injury severity, injury type, admission GCS, SBP, head AIS ≥3, and ventilator days	6
Shih et al. [21]	China, population-based	Case-control	2000–2018	General population	ATC classification system	Benzodiazepine	ICD-9-CM or ICD-10-CM	Case 12,546 Control 50,184	Hypertension, hyperlipidemia, chronic liver disease, chronic kidney disease, diabetes, chronic obstructive pulmonary disease,	9

(continued on next page)

Table 1 (continued)

Study, year	Location, setting	Study design	Study period	Population	Exposure measurement	Type of BZRD	Definition of DVT	Total population	Adjustment	Quality
Zhang et al. [19],	China, hospital-based	Case-control	2020	Patient admitted psychiatric units	Medical record review	Benzodiazepine	ICD-10	Case 41 Control 41	rheumatoid arthritis, ankylosing spondylitis, ischemic heart disease, stroke, varicose veins of lower extremities, anxiety disorder, insomnia, antiplatelet, and anticoagulant. NA	5

ATC, Anatomical Therapeutic Chemical; AIS, Abbreviated Injury Scale; BMI, Body Mass Index; BZRD, benzodiazepines and related drugs; GCS, Glasgow Coma Scale; GERD, gastroesophageal reflux disease; NA, not available; SBP, admission systolic blood pressure; VTE, Venous thromboembolism.

Table 2
Meta-analysis for studies included in the analysis.

Subgroup analysis	Number of studies	Number of estimates	VTE events (n)	Total subjects (n)	Pooled OR (95 % CI), I ² statistics (%), P-value for the heterogeneity Q test	Model used
Overall-analysis	8	10	18,997	545,985	1.42 (1.19–1.7); I ² = 88 %, P < 0.001	Random effects
Study quality						
High quality	4	5	15,748	545,159	1.47 (1.14–1.9); I ² = 91 %, P < 0.001	Random effects
Low quality	4	5	3249	381,886	1.27 (1.1–1.45); I ² = 32.7 %, P = 0.203	Random effects
Study setting						
Population-based	5	7	18,796	545,159	1.38 (1.14–1.68); I ² = 91.7 %, P < 0.001	Random effects
Hospital-based	3	3	201	826	1.72 (0.99–3.01); I ² = 32.2 %, P = 0.229	Random effects
Study design						
Cohort	4	5	3129	177,477	1.23 (1.11–1.36); I ² = 13.3 %, P = 0.329	Random effects
Case-control	4	5	15,868	368,508	1.53 (1.19–1.97); I ² = 90.8 %, P < 0.001	Random effects
No. of adjustment variables						
< 5	5	7	18,796	545,159	1.38 (1.14–1.68); I ² = 91.7 %, P < 0.001	Random effects
≥ 5	3	3	201	826	1.72 (0.99–3.01); I ² = 32.2 %, P = 0.229	Random effects
Studies adjusted for indication	3	4	15,411	82,076	1.42 (1.11–1.84); I ² = 92.9 %, P < 0.001	Random effects
BZRD type						
BZD	6	8	NA	NA	1.56 (1.26–1.92); I ² = 85.41 %, P < 0.001	Random effects
Z-drug	2	3	NA	NA	1.33 (1.05–1.68); I ² = 0 %, P = 0.405	Random effects

4. Discussion

To our knowledge, this is the first meta-analysis to evaluate the association between BZDR exposure and the risk of VTE. Through the rigorous aggregation of data from available studies, our findings demonstrated a significant association between BZDR utilization and an elevated risk of VTE; this was consistently supported by robust sensitivity and pre-specified subgroup analyses. However, the pooled OR of the hospital-based studies was high (OR = 1.72) but fall just short of statistical significance. Due to the limited power of the included hospital-based studies, there is no reason to assume the association doesn't hold for these populations. The focus of this analysis shouldn't be only on *p*-values and the point estimate needs to be taken into consideration.

The mechanisms underlying the association between BZDR use and increased VTE risk remain unclear; however, several potential pathways

may be involved. First, the sedative and muscle relaxant effects of BZDRs may reduce physical activity, leading to venous stasis in the lower limbs [28]. Second, benzodiazepines have been reported to cause blood pressure fluctuations, resulting in circulatory instability and impaired venous return [29]. Third, benzodiazepines can suppress respiratory function, leading to hypoxemia, which in turn may activate the coagulation system, increase haematocrit and blood viscosity, and impair endothelial function [30], ultimately contributing to an elevated VTE risk. Consistent with these biologically plausible mechanisms, our pooled analysis demonstrated a significant association between BZDR exposure and increased VTE risk. However, confidence in these findings is tempered by the significant heterogeneity across studies. Subgroup analyses based on study design, geographical location, study quality, clinical setting, and the extent of covariate adjustment failed to fully account for this heterogeneity. Considering the presence of clinical heterogeneity, some degree of statistical heterogeneity in our findings

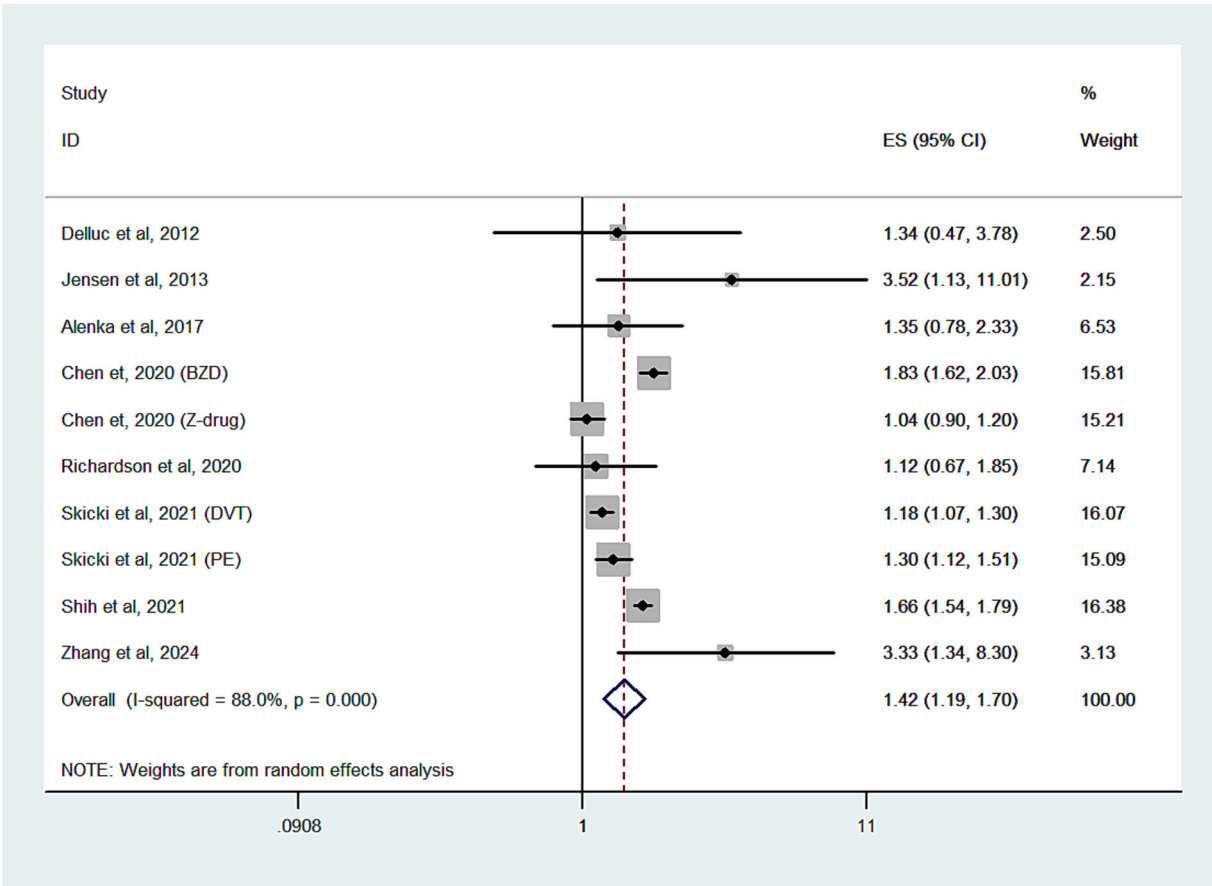


Fig. 2. OR and 95 %CI of VTE risk compared with BZRD users to non-users.

was anticipated.

Insomnia, a prevalent symptom of various psychiatric disorders, has been independently associated with an elevated VTE risk [31]; meanwhile, psychotic and bipolar disorders were also significantly associated with VTE risk [32]. Therefore, studies investigating this association should account for drug indications. The optimal approach to control for confounding by indication is to use a reference group of patients with established BZDR indications who did not receive the medication; however, none of included study met this standard. Although a positive association was observed in our overall analysis, the possibility of overestimation cannot be ruled out because five studies did not adjust for this confounding. Then, further analysis limited to three studies [20,21,23] explicitly control for drug indication was conducted and the pooled OR was not significantly altered, suggesting a true association exist.

Compared to patients with psychiatric disorders, surgical patients may be more susceptible to drug toxicity due to concurrent administration of anaesthetic agents, potentially diminished drug metabolism, and reduced clearance. Therefore, BZDR exposure in surgical patients could theoretically increase VTE risk. However, our findings revealed the VTE risk was lowest in the surgical patient subgroup. This may be attributed to the intensive care provided to surgical inpatients, including routine administration of prophylactic anticoagulants, which could mitigate the VTE risk associated with BZDR use. However, our findings are limited by the small sample size. Further studies are needed to validate these findings.

BZDRs may increase VTE risk through their sedative and muscle relaxant effects, indicating a potential dose-dependent relationship. Among the included studies, Chen et al. [20] observed a defined daily dose effect, wherein increased cumulative BZDR exposure was associated with an elevated VTE risk. Similarly, Shih et al. [21] demonstrated

a high VTE risk associated with increased cumulative BZDR dose. However, no studies have directly compared the risk across low, medium, and high doses. Further studies are needed to elucidate the dose–response relationship between BZDR use and VTE risk.

Current evidence suggests generally moderate risk of VTE. However, our findings might not support the discontinuation of BZDRs, as they are an effective intervention for some major psychiatric conditions. Given the large number of BZDR users, clinicians should ensure that BZDRs are prescribed only for patients with a clear indication and be cautious when prescribing BZDRs to subjects who have an underlying VTE risk. Furthermore, these patients prescribed these medications should be monitored during the course of treatment if VTE-associated signs and symptoms appear, considering the possibility of treatment withdrawal.

To our knowledge, this is the first systematic review and meta-analysis to evaluate the effect of BZDR exposure on VTE risk. The primary strengths of our study include a comprehensive literature search and a rigorous systematic review and meta-analysis of all relevant studies to date. In addition, we conducted multiple stratified and sensitivity analyses to assess the robustness of our findings. However, our study has several limitations. First, the inherent limitations of epidemiological observational studies, particularly the potential for residual confounding, must be acknowledged. Several included studies did not adequately adjust for known VTE risk factors, impacting the robustness of our conclusions. Second, the presence of substantial heterogeneity necessitates a cautious interpretation of the meta-analysis results. Although subgroup analyses were performed, they failed to fully account for this heterogeneity. Third, the extraction of detailed information regarding BZDR dosage and specific drug types was precluded by limitations in the reported data. Therefore, the potential influence of these critical exposure parameters on VTE risk could not be definitively assessed.

In conclusion, this systematic review and meta-analysis suggests an association between BZDR use and increased VTE risk. In light of our study's limitations, large-scale, well designed studies that consider greater numbers of confounding factors and BZDR indications are needed to confirm our findings.

Details of ethical approval

No ethical approval was required for this review as all data were already published in peer-reviewed journals.

CRediT authorship contribution statement

Wei Yang: Validation, Project administration, Conceptualization. **Zi-hao Wang:** Supervision, Software, Methodology, Formal analysis, Data curation. **Ye-ke Zhang:** Supervision, Software, Resources, Data curation. **Xiao-feng Xu:** Visualization, Software, Resources, Investigation. **Hong Dai:** Methodology. **Hai-yin Jiang:** Methodology, Investigation. **Ji-yuan Ding:** Writing – review & editing, Writing – original draft.

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Declaration of competing interest

No specific funding for the purpose of this meta-analysis was received from any bodies in the public, commercial, or not-for-profit sectors.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.genhosppsych.2025.10.018>.

Data availability statement

The data used in this study are openly available on online databases. Further inquiries can be directed to the corresponding authors.

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